

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented production-grade agentic systems in **Python** for mission-critical, regulated workflows, focusing on stateful orchestration.
- Developed agent workflows using **LangGraph**, managing multi-step execution and implementing context management strategies for long-running processes.
- Built and maintained **RAG pipelines**, integrating structured, semi-structured, and unstructured data sources with advanced retrieval strategies.
- Engineered **Python** data pipelines and ETL processes for automated reporting and data transformation tasks, processing approximately **2 TB** of data weekly.
- Developed **Java Spring Boot** REST APIs for cross-service communication, ensuring robust error handling and efficient data exchange patterns.
- Containerized applications using **Docker** and managed deployments on **AWS EC2** instances, integrating with **Jenkins** for CI/CD pipelines.
- Provided production support for enterprise applications, responding to P1/P2 incidents and managing SLAs using **Splunk** and **Dynatrace**.

SKILLS

AI/ML – AI, Agentic Systems, RAG, Embeddings, Retrieval Strategies, Model Serving, Python

Frameworks/Libraries – LangGraph, LangChain, Scikit-learn, Pandas, NumPy

Languages – Python, Java, SQL

Databases – PostgreSQL, MySQL, Oracle, NoSQL, Vector Stores

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins

Methodologies – Agile, Scrum, Workflow Orchestration, State Management, Retries, Idempotency

PROJECTS

AI Agentic System for Workflow Automation

- Developed a Python-based agentic system to automate complex financial reconciliation workflows, reducing manual processing time by **40%**.
- Implemented a RAG pipeline to ingest and query diverse transaction data sources, enabling intelligent decision-making within the agent's operational logic.
- Designed state management and retry mechanisms using LangGraph for reliable, long-running agent executions in a simulated regulated environment.

Intelligent Data Ingestion Pipeline

- Built a Python ETL pipeline to process and structure semi-structured and unstructured financial documents for an AI analytics platform.
- Integrated vector embeddings and retrieval strategies to enable efficient searching and context retrieval from a large corpus of internal documentation.
- Deployed the pipeline on AWS Lambda, handling an average of **1,500** requests per hour with a p95 latency of **220ms**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)